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Motor Fault Prediction Based on Graph Convolutional Networks with Signal Association Mining

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Abstract

Vibration signals are often used to analyze and recognize the characters of faults that verify in the operation of induction motors. At present, the processing method of fault data is mainly through expert judgment and analysis, and the processing cost is high. In recent years, induction motor fault data processing models with deep learning method have been utilized in induction motor fault diagnosis tasks. In the study, a type of deep learning algorithm of graph convolutional neural network based on multi-head attention enhancement is proposed to determine the fault type of induction motor by using time series fault data. The proposed method firstly obtains the extended time sequence feature representation, encodes the input fault node data through utilizing graph convolutional network, and finally realizes the obtained fault type classification of induction motor through recurrent neural network and nonlinear layer. The corresponding experimental results shown that the proposed approach could achieve the competitive performance in certain dataset.

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Keywords: Vibration signals, Motors; Fault diagnosis, Graph convolutional Network, Recurrent neural network

1. Introduction

Electric motors are critical equipment in power systems. Once a fault or shutdown occurs, it will not only damage the motor itself but also lead to huge economic losses and even potential casualties. Therefore, the early prediction and diagnosis of potential motor faults are of great significance [1-2]. At present, motor fault diagnosis approaches could be classified into two categories: traditional diagnosis approaches and the methods with utilizing deep learning algorithms. Traditional fault diagnosis methods rely on manual extraction of effective signal features from recorded motor operation data, and then use shallow machine learning algorithms such as support vector machines, random

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forests, and fuzzy inference to learn the operation and fault modes from manually constructed features, so as to realize motor fault classification and diagnosis [3]. Traditionally, motor fault diagnosis has been regarded as a knowledge-driven discipline [4-5]. It requires researchers to possess sufficient professional domain knowledge—including expertise in signal processing, vibration mechanics, and fault mechanisms—to manually design and construct effective features [6]. In the feature processing stage, researchers need sufficient professional domain knowledge for feature construction, and the time cost is relatively high [7-9]. With the rapid development of sensor method, the detectable motor operation signals have shown a trend of large volume, diversity, and high speed in both accuracy and signal types [10]. Of course, we note that the previously proposed corresponding algorithms that might only consider the analysis model or certain algorithms on certain specific problems such as population migration and disease transmission [11]. Traditional manual feature design and extraction methods have become extremely difficult, and the corresponding time and labor costs are also unbearable [12]. It seems a small real world. To be specific, in a social network, adjacency matrix is an important representation method for a graph. However, adjacency matrix only keeps the local information of network node structure and the citation network node structure information of non-adjacent nodes cannot be kept.

By constructing a graph where nodes represent sensor signals or extracted features and edges denote their physical or statistical relationships, the GCN can explicitly model the non-Euclidean structure of the data [13]. Furthermore, the multi-head attention mechanism enables the method to comprehensively obtain to node information from the various dimension representation node subspaces at different locations, capturing nuanced cross-channel dependencies and yielding more refined temporal encodings than conventional approaches. In AGM algorithm, a social network is divided into overlapping part and non-overlapping part, which can be attributed a larger edge connection density than the non-overlapping part; therefore a probabilistic node graph structure is emerged to the structure and the nodes can be partitioned into communities by using different network node layer; MDDBMS employs the degree of each node to evaluate the density of each node, then the node with largest density is selected as the initial node of a cluster; the cluster is extended by neighbourhood search and all nodes which belong to the current cluster are added into the cluster by the discriminant criterion [14].

Firstly, the multi-head attention mechanism maps low-dimensional signal features into different embedding spaces, and the enhanced signals of the final time-series nodes are obtained through signal fusion [15]. For the encoded enhanced signals of different time nodes, the cosine similarity scores between every two time-series node embeddings are calculated and sorted. The Top-K potentially correlated node pairs are selected to construct an adjacency matrix with 0/1 weights. However, notwithstanding the evaluate methods above obtained competitive performance, they only consider the value of different link weightings between different subspace layers, which can be cause an effect on evaluation experimental results. Inspired by the MOEE model framework, the graph node network structure can be utilized to encode the time-series nodes to achieve motor structure operation node information. Finally, the LSTM network and fully connected layers are adopted to achieve context learning of time-series data and classification of the motor operation state at the current time node, namely the classification of motor fault types. The corresponding experimental results shown that the proposed method achieved the competitive performance in terms of high-precision fault type prediction, which can be used to motor operation monitoring positively.

2. Model Design for Motor Fault Diagnosis Analysis

Motor fault signals are usually single time-series vibration signals, and the cost of manual data processing and analysis is high. To tackle the problems above, we propose an approach based on graph convolutional network enhanced by the multi-head attention mechanism (GCN-MHAM) to mine and obtain the original fault vibration signals of motors. As illustrated in Fig. 1, the overall framework comprises four main modules:

(1) Multi-head attention layer: this multi-head attention framework [9] can be utilized to map the original differential fault signals into different vector spaces. After vector concatenation, the original fault feature signals are extended to higher feature dimensions. Also, the algorithm uses conductivity and modularity to discover the nodes in the same cluster; the algorithm selects two nodes with maximum edge weight as an initial cluster; then it selects a node from neighborhood nodes with maximum edge weight.

(2) Graph convolutional neural network layer: this graph convolutional layer calculates the corresponding subspace cosine similarity between the extended node vector value at different time instants to obtain similarity scores. The Top-K feature vector pairs are selected as node pairs with potential edge connections in the graph, so as to construct an adjacency matrix composed of 0/1 weights. Through the aggregation operation between nodes, feature encoding among nodes at different distances, especially long-range nodes in the time series, is realized. To be more specific, the network structure node relevance between two different layers are primarily affected by two analysis elements, namely, structure similarity (relation-based) and interaction layer structure (shape-based). Assume that the network node relevance between two different interaction layers is excellent whether the corresponding nodes could achieve more topology value of the node structure.

(3) Sequential learning module: The LSTM network is introduced to further learn temporal features in the sequence, and finally a comprehensive temporal representation within a certain range is obtained.

(4) Classifier network: An effective prediction of fault types is achieved by using a classification activation function based on Softmax.

The detailed functions and formulas of each module are given below.

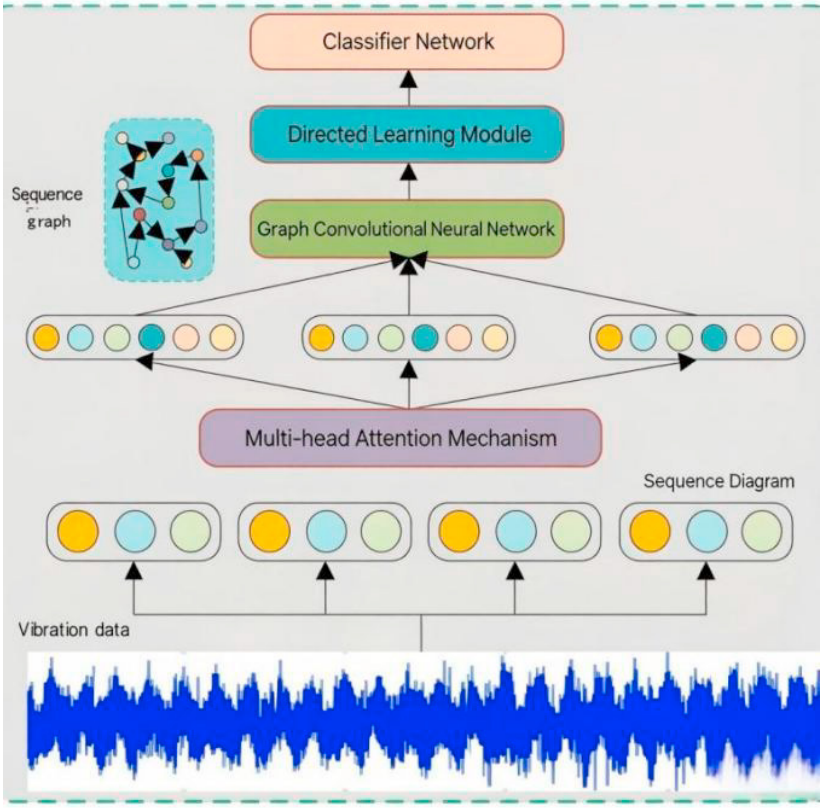


Fig. 1. GCN-MHAM model structure diagram

Since the original vibration signals are a series of discrete signals at each time step, this paper adopts the differential processing method to construct a basic mapping vector for the vibration signal at each time step in the time series. The signal after differential processing at each time step is given by the Equation (1):

$$S_t = [s_t - s_{t-1}, s_t, s_{t+1} - s_t] \in \mathbb{R}^3 \tag{1}$$

in which $1 \le t \le T\text{-}1$, T denotes the temporal length of a segment of vibration signal input into the model. After obtaining the initial features at time t, the multi-head attention mechanism can be introduced to the map constructed

initial feature signals into different vector spaces, respectively. Of course, for the *i*-th projection element head, the mapped representation obtained after the attention mechanism is given by Equation (2):

$$\begin{cases} Q_t = W_q \cdot S_t, K_t = W_K \cdot S_t, V_t = W_v \cdot S_t \\ x_t^i = softmax\left(\frac{Q_t \times K_t^T}{\sqrt{3}}\right) V_t \in R^{d_k} \end{cases} \tag{2}$$

in which $W_q \in \mathbb{R}^{3\times d_q}$, $W_k \in \mathbb{R}^{3\times d_k}$ and $W_v \in \mathbb{R}^{3\times d_v}$ are all projection matrices. The final feature vector obtained by multi-head and concatenation is $\overline{x_t} = [x_t^1, x_t^2, \ldots, x_t^n] \in \mathbb{R}^d$, $d=nd_v$. In addition, there are much redundant information in adjacency matrix because adjacency matrix is a symmetric matrix. Thus, this study proposed an approaches which can utilize and obtain the global structure information of nodes and transform the corresponding advanced matrix into the node network with higher weight value.

Based on the concatenated vector obtained above, the cosine similarity between every two nodes on the time series is calculated to obtain the similarity ranking among nodes, so that the potential correlations between different nodes can be established. The Top-K node pairs in the ranking score are selected as the node tuples in the graph construction, thereby constructing a binary adjacency matrix composed of 0/1 weights on the time series with length $TA \in \mathbb{R}^{T\times T}$, according to the obtained node representations $\bar{x}_T = [\bar{x}_0, \bar{x}_1, \ldots, \bar{x}_T] \in \mathbb{R}^{T\times d}$, the convolution operation performed by the model in the *l*-th layer of the Graph Convolutional Network (GCN) is shown in Equation (3):

$$x_T^{(g,l)} = \sigma\left(\tilde{D}^{\frac{l}{2}}\tilde{A}\tilde{D}^{\frac{l}{2}}\bar{x}_T^{(g,l-1)} W_g^{(l)}\right) \tag{3}$$

in which $\tilde{A} = A + I$, $\tilde{D}$ denotes degree matrix value of $\tilde{A}$, $W_g^{(l)} \in \mathbb{R}^{d\times d}$ denotes the mapping vector, $\sigma(.)$denotes the sigmoid activation function. After encoding through *L* layers of GCN, the model feeds the obtained embeddings into an LSTM network in the directed learning module to capture the contextual information along the sequence, as shown in Equation (4).

$$x_{final} = \overrightarrow{LSTM}(x_{t-1}^{(g,L)}, x_t^{(g,L)}) \tag{4}$$

Finally, the hidden states x_{final} within the learned time series *T* are fed into a nonlinear layer for classification, and the classification formula is given by Equation (5).

$$\hat{y} = softmax(W_{final} \cdot x_{final} + b_{final}) \in \mathbb{R}^{\delta} \tag{5}$$

3. Experiments and Analysis

To validate the effectiveness of the proposed method, we conduct a series experiments on a publicly available motor fault diagnosis dataset [10]. The dataset comprises time-series vibration signals acquired under eight different operating conditions: one healthy state and seven faulty states (including bearing defects, unbalanced rotors, and broken rotor bars). Each signal is collected at a specified sampling frequency value of 12 kHz. Furthermore, the dataset contains eight types of motor operating conditions, including the normal working condition. Furthermore, during the training process, the optimal hyperparameters obtained by the model through grid search are listed in Tab. 1.

It can be found from Tab. 2 that the corresponding comparison experimental results between the proposed study and the other methods such as as Convolutional Neural Network (CNN) model between the Long Short-Original-Term Memory (LSOTM) mechanism. These results validate the advantage of integrating multi-head attention with sequential modeling for this task. Non-negative matrix factorization is an effective method to decrease the dimension of data and keeps the good explanation of decomposition result. This study utilize the higher value advanced matrix factorization to achieve the node relation structure due to the more complex topology network structure.

Table 1. Hyperparameter Settings of the Proposed Model

Parameter	Hyperparameter value
Training Epochs	100
T	32
Learning rate	1e-3
GCN layers	2
d_q, d_k, d_v	{3,3,5}

It can be observed from Tab. 2 that the simple LSTM-based method can barely extract effective information from the constructed differential features. This is mainly because the input differential signals are low-dimensional features, making it difficult for LSTM to learn representative features directly from abstract time-series signals.

By incorporating CNN into the model and combining it with attention-weighted computation, the model is more advanced at to adaptively focus on critical node network information within the input batch data. We note that the integration and development of the attention mechanism approach, the model performance is significantly improved. Moreover, corresponding the results of the specific model and the corresponding algorithms on the obtained data sets. From the results, it is known that GGE-NMF algorithm and NMF algorithm obtain 2 best results and MDDBMS obtains 1 best result. The mean modularity of GGE-NMF algorithm is the mean value of all test objects. The corresponding results demonstrated that GGE-NMF framework can be viewed an competitive method for addressing network node discovery. On dolphins and football data sets, the performance of k-means algorithm and NMF algorithm is better than GGE-NMF algorithm and the advantages of k-means algorithm and NMF algorithm are obvious. For dolphins and football data set, the sparsity degrees of the two data sets are larger than the other data sets and it means that dolphins and football data sets are dense network

Nevertheless, the aforementioned approaches still fail to capture the interdependencies among temporal features at different time steps. For this reason, this paper introduces the GCN module, which enables the model to achieve the highest fault recognition accuracy. After the clusters of nodes are obtained, an adjustment method theory is interacted to revise and optimize the final network node and interaction layer result. To achieve the simplify of the framework, we only adjust partition of nodes and does not introduce the mechanism of merging communities.

Furthermore, this paper investigates the influence of the number and structure of GCN layers on the method performance test, the experimental results can be found in Fig. 2.

Table 2. Precision scores of models with different configurations

Model	Accuracy (%)
LSTM	36.54
Attention + LSTM	70.21
CNN+LSTM	85.46
Attention + CNN	92.22
GCN-MHAM	**95.77**

It is clear in Fig. 2 that the model performance does not increase monotonically with the variation in the number of GCN layers. Instead, due to the over-smoothing problem of GCN [11], it first rises and then declines. Therefore, in the experimental setup, the number and the specific network structure of GCN layers should not be set too large; otherwise, it will risk degrading model performance.

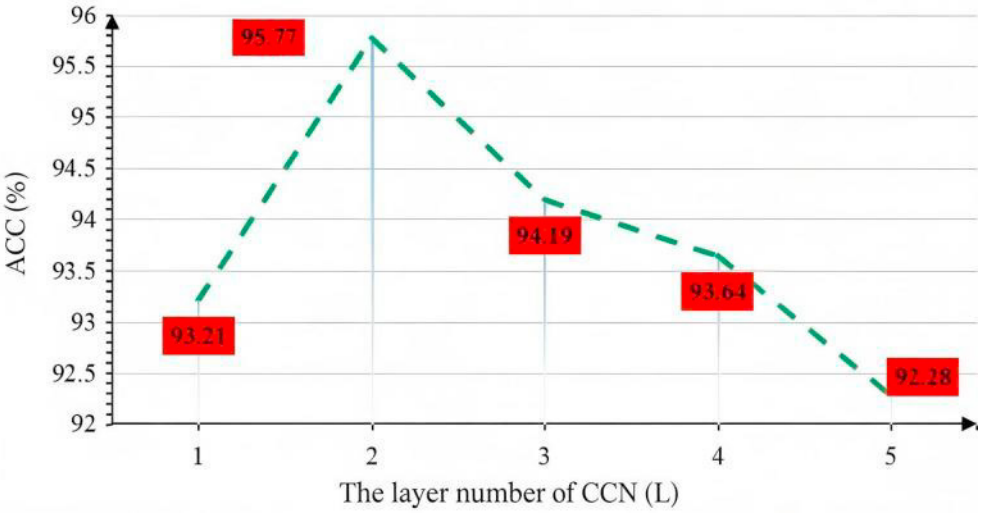


Fig. 2. Model performance with variations in the number of GCN layers

4. Conclusion

In this study, a fault motor classification model based on multi-head attention-enhanced graph convolutional neural network is proposed. After differential processing on the data, the adopted multi-head attention mechanism can obtain high-dimensional signal feature representations from low-dimensional features through multi-head mapping. The graph convolutional network is then used to perform graph construction and node aggregation operations to achieve better temporal feature encoding, thereby improving the performance of motor fault classification. In the future, based on the above ideas, this paper will focus on the over-smoothing problem in graph convolutional networks, so as to improve the prediction performance of the model when a large number of graph convolutional layers are employed. Moreover, we would like to verify the topology structure of the network structure layer and specific number. For instance, the effect of proposed model framework might lead to the interaction of the corresponding entity layers. Actually, the dynamic and variation can be evaluated certain other evaluation metrics such as TTSC and TOSST. Of course, we note that a series factor selection methods are mainly based on the network topology structure.

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